MathQuest | Expert Review Pack

Learner Profile & Prediction Engine • Draft for review • 17 October 2026

Reviewer: Ms. Pathmasri Sivalingam (Clinical Psychologist) Meeting notes by: _____

Purpose: Review the plausibility, child-appropriateness and safeguards of preliminary rules. These rules are hypotheses for expert review, NOT clinically validated thresholds, diagnoses or final scoring logic.

A. Five-minute introduction

MathQuest is an adaptive Sinhala mathematics learning game for children aged 6–10. My component combines a short baseline activity and subsequent gameplay evidence to describe what the learner can currently do, where help is useful, and whether skills carry over to later independent tasks. It supports educational decisions; it does not detect, classify or diagnose ADHD.

Possible evidence (subject to feasibility and review): correctness on grade-aligned tasks, number of attempts, hint use, task completion, performance on a comparable independent task, and changes across sessions. Task difficulty and accessibility must be considered before interpreting these signals.

B. Draft decision rules — request accept / revise / reject / not within expertise

R1 Do not label a mathematics concept as mastered from one correct response. Look for performance across multiple comparable opportunities.

Ask expert: How much evidence is enough? What exceptions matter?

Decision: ☐ Accept ☐ Revise ☐ Reject ☐ Outside expertise Notes: _____

R2 When a child struggles repeatedly with a concept, suggest an easier worked example or a smaller step rather than simply repeating the same item.

Ask expert: What supportive intervention is appropriate?

Decision: ☐ Accept ☐ Revise ☐ Reject ☐ Outside expertise Notes: _____

R3 If the child succeeds after a hint, record supported success separately from independent success; do not automatically treat them as the same.

Ask expert: Is this distinction educationally fair?

Decision: ☐ Accept ☐ Revise ☐ Reject ☐ Outside expertise Notes: _____

R4 After supported success, offer a comparable unassisted opportunity later before inferring independent performance.

Ask expert: What delay and format would be suitable?

Decision: ☐ Accept ☐ Revise ☐ Reject ☐ Outside expertise Notes: _____

R5 If responses are unusually slow or incomplete, do NOT infer inattention or ADHD; allow for reading, motor, device, language, fatigue and task difficulty factors.

Ask expert: What alternative explanations must be captured?

Decision: ☐ Accept ☐ Revise ☐ Reject ☐ Outside expertise Notes: _____

R6 If repeated attempts show frustration or disengagement signals, offer an optional break/help/shorter activity without penalty.

Ask expert: Which observable signs justify support, and how to avoid false positives?

Decision: ☐ Accept ☐ Revise ☐ Reject ☐ Outside expertise Notes: _____

R7 Where evidence is sparse, inconsistent or disrupted, report "insufficient evidence" rather than a fixed learner category.

Ask expert: What minimum evidence/confidence rules are sensible?

Decision: ☐ Accept ☐ Revise ☐ Reject ☐ Outside expertise Notes: _____

R8 Describe parent/teacher outputs as observable patterns and suggested support, not clinical traits or fixed ability labels.

Ask expert: What language is safe and understandable?

Decision: ☐ Accept ☐ Revise ☐ Reject ☐ Outside expertise Notes: _____

C. Practical learner scenarios

Case 1 — Hint-dependent success

A child misses two subtraction items, answers correctly after a worked hint, then misses a similar item without a hint.

Discuss: Would "needs more guided practice" be a fair temporary interpretation? What else should we test?

Case 2 — Slow but accurate

A child solves four age-appropriate items correctly but takes longer than peers.

Discuss: Should duration alone change support level? What confounding factors matter?

Case 3 — Uncertain pattern

A child answers one item correctly and one incorrectly, then stops because of a device interruption.

Discuss: What should be shown when evidence is insufficient?

D. Reviewer questions

1. Which indicators are suitable, unsuitable or missing for 6–10-year-old children?

2. Which rules could unfairly label a child or confuse symptoms with learning needs?

3. Should rules differ by age, grade, task complexity or support requirements?

4. Which recommendations are helpful, neutral or potentially harmful?

5. Should we review any questionnaires (including SNAP-IV) separately from in-game mathematics decisions?

6. What wording and safeguards are needed for parent/teacher dashboards?

7. What additional educational or clinical experts should review the rules?

E. Validation record

Rule ID	Accept / revise / reject / outside expertise	Reason / proposed change
R1	_____	_____
R2	_____	_____
R3	_____	_____
R4	_____	_____
R5	_____	_____
R6	_____	_____
R7	_____	_____
R8	_____	_____

Overall expert comments: _____

Additional evidence or literature requested: _____

Follow-up action / owner / date: _____

Reviewer name: _____ Meeting date: _____ Review method: ☐ Discussion ☐ Written review

Note: A discussion supports preliminary expert feedback. A formal expert-validation claim requires documented item-level judgments, revisions, and a defined protocol (and, where appropriate, multiple reviewers).